CSE-423 Computer Graphics

Topics	To be discussed after Midterm
Transformations/ motion	Geometric and coordinate transformation. Homogeneous Coordinate system, 4x4 translation matrix. Derivation of 3D rotation matrix for center of rotation (0, 0, 0) and (a, b, c). Composition of geometric transformations. Rotation matrix-based problems.
Projection	Definition, Classification. Derivation of simple perspective projection matrices and general-purpose projection matrix. Simplification of general-purpose projection matrix. General-purpose projection matrix-based problems.
Color model of light	Comparison between additive and subtractive color. RGB color model and CMY color model. HSV and HLS color model. RGB-to-HSV and RGB-to-HSL Conversions algorithm. Problem on RGB-to-HSV and RGB-to-HSL conversions.
Illumination and Shading	Illumination/light model, Global and local light model. Ambient light. Diffuse and specular reflection model. The Phong reflection/illumination model. Constant/flat shading and interpolated/smooth shading. Phong-shading vs Gouraud-shading,
Curve and surfaces	Standard equation of Cubic Polynomial, Types of curves. Continuity of curve segments. Derivation of basis-matrix and blending functions of Hermite-curve and Bezier-curve. Basis-matrix based problems.